Angles — Practice — Answer Key

A worked example comes before each type. Show your thinking.

Name: _____ Date: _____

Type 1 Classify the Angle

Watch Me — Worked Example

Classify an angle of $132°$.
It is more than $90°$ but less than $180°$. **Answer: obtuse**

1. $47°$

acute

2. $90°$

right

3. $179°$

obtuse

4. $180°$

straight

Type 2 Angle Addition Postulate

Watch Me — Worked Example

D is inside $\angle ABC$. $m\angle ABD = 28°$ and $m\angle DBC = 47°$. Find $m\angle ABC$.
Part + part = whole: $28 + 47 = 75$. **Answer: 75°**

5. $m\angle ABD = 40°$, $m\angle DBC = 65°$. Find $m\angle ABC$.

$105°$

6. $m\angle ABC = 120°$, $m\angle ABD = 45°$. Find $m\angle DBC$.

$75°$

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7. $m\angle ABD = x + 10$, $m\angle DBC = 2x + 5$, $m\angle ABC = 90°$. Find x and $m\angle DBC$.

$x = 25,\ 55°$

8. HARD $\overrightarrow{BD}$ bisects $\angle ABC$. $m\angle ABD = 3x + 4$, $m\angle DBC = 5x - 12$. Find $m\angle ABC$.

$x = 8,\ 56°$

Type 3 Complements & Supplements

Watch Me — Worked Example

Find the complement and the supplement of $38°$.
Complement: $90 - 38 = 52°$. Supplement: $180 - 38 = 142°$. **Answer: $52°$, $142°$**

9. Find the complement of $71°$.

$19°$

10. Find the supplement of $98°$.

$82°$

11. Two angles are complementary. One is twice the other. Find both.

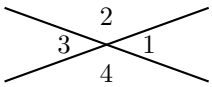
$30°,\ 60°$

12. Two angles are supplementary: x and $3x - 20$. Find both.

$50°,\ 130°$

Type 4 Vertical Angles & Linear Pairs

Watch Me — Worked Example



$m\angle 1 = 65°$. Find $m\angle 3$ and $m\angle 2$.
Vertical angles are equal: $m\angle 3 = 65°$. Linear pair: $m\angle 2 = 180 - 65 = 115°$.
Answer: $65°$, $115°$

Use the picture in the example for problems 13 and 14.

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13. $m\angle 2 = 128°$. Find $m\angle 4$.

$128°$

14. $m\angle 1 = 73°$. Find $m\angle 2$.

$107°$

15. Vertical angles measure $5x - 10$ and $3x + 30$. Find x and the angle.

$x = 20,\ 90°$

16. A linear pair measures $2x + 15$ and $3x - 5$. Find both angles.

$83°,\ 97°$

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